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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 02/03/2006 |
| **PI Name:** | Sangani, Ashok S. Ph.D. | | |
| **Project Title:** | Behavior of rapidly sheared bubble suspensions | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Dynamics and stability | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Syracuse | **State:** | NY |
| **Zip Code:** | 13244-1190 | **Congressional District:** | 25 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | OTHER |
| **Start Date:** | 09/01/1996 | **End Date:** | 08/31/2006 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 2 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Nahra, Henry K | **Contact Phone:** | 216-433-5385 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Koch, Donald L ( Cornell University ) | | |
| **Grant/Contract No.:** | NAG3-2427 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | The objective of this experiment is to test the equations of motion of bubble suspensions in an ideal setting where potential flow prevails, and thereby validate the concepts of bubble pressure and effective viscosity of bubble suspensions. The conditions that promote potential flow are that the Reynolds number be large and the Weber number be small. The means to achieve the objectives are to generate a bubble suspension in a Couette cell in the microgravity environment and to shear it by rotating the outer cylinder.The forces acting on the bubbles in this configuration are the centrifugal force which pushes the bubbles toward the inner cylinder and the shear induced bubble phase pressure which tends to push the bubbles apart from each other. To accomplish the objective, measurements of the volume fraction, and the liquid and bubble velocity will be taken as functions of radial position. In addition the shear stress at the wall will be measured. Ground-based research objectives are: (i) to produce nearly monodisperse suspensions of nearly spherical, non-coalescing bubbles; (ii) to develop probes for measuring volume fraction and velocity distributions of the bubbles as a function of position; (iii) to measure wall shear stress; (iv) to perform experiments for bubble suspensions in vertical | | |

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| | and inclined channels and in circular, vertical pipe configurations; and (v) to continue analytical work related to the description of flows of bubble suspensions. Multphase flows occur in many industrial processes. The questions concerning the stability of well-dispersed multiphase medium and its properties are important in predicting the behavior of multiphase flows and in designing equipment in which these flows occur. The detailed investigation of the idealized bubble suspensions under consideration is a first step toward developing an analytical framework for answering these questions. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Bubbly liquids are encountered in nature and in many industrial processes. These processes can be designed better if the properties of bubbly liquids and their equations of motion are known. Understanding forces responsible for seggregation and clustering and coalescence is crucial for designing reliable equipment, e.g. bubble reactors and gas-liquid transport lines, that handle gas-liquid flows. |
| **Task Progress:** | Study of flows of bubbly liquids through a vertical pipe was completed. Bubble volume fraction, liquid velocity, and bubble velocity, and the velocity variance profiles across the pipe cross sections were determined for different overall bubble volume fractions and Reynolds numbers. The measured values were compared with those predicted by a theory based on averaged equations and very good agreement between the two was observed. Preliminary results are reported in Sangani et al. (2005). A manuscript detailing the experiments and comparison with the experiments is currently in preparation.<br>A lattice-Boltzmann method was developed for simulating flows of liquid containing nearly spherical and non-coalescing bubbles. The method directly recovers the slip (vanishing tangential stress) boundary condition so one only needs to solve the flow in the continuous, liquid phase. Compared with two-component numerical methods, the new single-component is faster and more robust. With the new method we were able to simulate and study many of the fundamental properties of a homogeneous bubbly suspension, such as hindered rising velocity, velocity variance and microstructures. For vanishing small Reynolds number case, our simulation results for the average rise velocity of the bubbles agreed well with the results available in the literature (Sangani and Acrivos, Int. J. Multiphase Flow, 1982). We have used the method to simulate sheared bubble suspension at Reynolds number based on bubble radius and overall shear rate of O(20). Calculations at other Reynolds numbers are underway. Bubble wall interactions lead velocity distribution of bubbles to be bimodal and anisotropic. A simple theory based on nearly isotropic velocity distribution with Newtonian closures are inadequate to describe flows in narrow channels. We have developed a kinetic theory that accounts for the bubbel-wall interactions and the resulting bimodal, anisotropic distribution. The results of the theory are in very good agreement with the numerical simulations. A manuscript based on this work is in preparation.<br><br>A. S. Sangani, S. S. Ozarkar, Y.H. Tsang and D. L. Koch, “Flow of bubbly liquids in a vertical pipe: Theory and Experiments,” Proc. IUTAM Symposium on Computational Approaches to Disperse Multiphase Flow (2005). |
| **Bibliography Type:** | Description: (Last Updated: 04/02/2007) |
| **Papers from Meeting Proceedings** | Sangani AS, Ozarkar SS, Tsang YH, Koch DL. "Flow of bubbly liquids in a vertical pipe: Theory and experiments." IUTAM Symposium held at Argonne National Laboratory, October 4–7, 2004.<br>Proceedings of an IUTAM Symposium held at Argonne National Laboratory, October 4–7, 2004, in press 2005. , Dec-2005 |